

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 31

WATER POLLUTION - 8

SOLID WASTE

INTRODUCTION

- Solid waste comprises of paper, food waste, plastics, glass, metals, rubbers, leather and textile etc.
- Landfills are adopted as the substitute for open burn dumps. In a landfill the land is dug up and solid waste is dumped and covered with mud and soil.
- Landfills are also not really much of a solution since the amount of garbage generation especially in the metros has increased so much that the landfill sites are getting filled too.
- Also, there is danger of seepage of chemicals, etc. from these landfills polluting the underground water resources.
- So, is an alternative solution sanitary landfill.
- 62 million tons of waste is generated annually in the country at present, out of which 5.6 million tons is plastic waste and 15 lakh tons is e-waste.
- Only about 75-80 per cent of the municipal waste gets collected and only 22-28 per cent of this waste is processed and treated.



SOLID WASTE MANAGEMENT RULES

The new rules are now applicable beyond municipal areas and have included census towns, notified industrial townships, areas under the control of Indian Railways, airports, special economic zones, places of pilgrimage, religious and historical importance, and State and Central Government organizations in their ambit.

I SEGREGATION AT SOURCE

- The new rules have mandated the source segregation of waste. Waste generators would now have to now segregate waste into three streams-
 - ✓ Biodegradables
 - ✓ Dry (Plastic, Paper, metal, Wood, etc.) and
 - ✓ Domestic Hazardous waste (diapers, napkins, mosquito repellents, cleaning agents etc.)before handing it over to the collector.
- In case of an event, or gathering of more than 100 persons at any place, the organizer will have to ensure segregation of waste at source and handing over of segregated waste to waste collector.



SOLID WASTE MANAGEMENT RULES

R U L E

I SEGREGATION AT SOURCE

- All hotels and restaurants will also be required to segregate biodegradable waste and set up a system of collection to ensure that such food waste is utilized for composting or for Biomethanation.
- Biomethanation is the process of converting organic material or waste into Biogas & Manure with the help of anaerobic microorganisms.
- Biogas is primarily Methane and CO_2 . Biogas can be produced from raw materials such as agricultural waste, municipal waste, plant material, sewage and food waste.

BIOMETHANATION

- 1 — ORGANIC MATTER
- 2 — COLLECTION
- 3 — BIOMETHANATION
- 4 — DISTRIBUTION N/W
- 5 — ENERGY



SOLID WASTE MANAGEMENT RULES

R U L E

II COLLECT BACK SCHEME FOR PACKAGING WASTE

As per the rules, brand owners who sell or market their products in packaging material which are non-biodegradable, should put in place a system to collect back the packaging waste.



SOLID WASTE MANAGEMENT RULES

R U L E

III USER FEES FOR COLLECTION

- Municipal authorities will levy user fees for collection, disposal and processing from bulk generators. As per the rules, the generator will have to pay “User Fee” to the waste collector and a “Spot Fine” for littering and non-segregation.
- Also, the new rules have mentioned about the integration of rag pickers from the informal sector to the formal sector by the state government.
- The rules also stipulate zero tolerance for throwing; burning, or burying the solid waste generated on streets, open public spaces outside the generator’s premises, or in the drain, or water bodies.

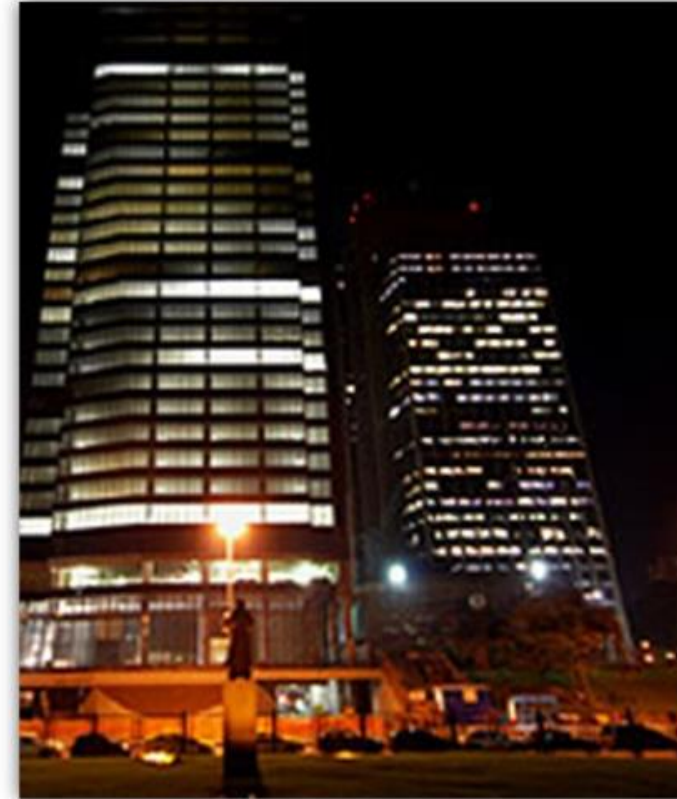


SOLID WASTE MANAGEMENT RULES

R U L E

IV USER FEES FOR COLLECTION

- The developers of Special Economic Zones, industrial estates, industrial parks etc., to earmark at least 5 per cent of the total area of the plot for recovery and recycling facility.
- Waste processing facilities will have to be set up by all local bodies having a population of 1 million or more within two years.



SOLID WASTE MANAGEMENT RULES

R U L E

V PROMOTING USE OF COMPOST

- As per the rules, the Department of Fertilizers, Ministry of Chemicals and Fertilizers should provide market development assistance for city compost.
- Also, the Ministry of Agriculture should provide flexibility in Fertilizer Control Order for manufacturing and sale of compost, to propagate use of compost on farm land, set up laboratories to test quality of compost produced by local authorities or their authorized agencies.



Ministry of Agriculture
& Farmers Welfare,
Government of India



Department Of Fertilizers
Ministry of Chemicals and Fertilizers
Government of India

SOLID WASTE MANAGEMENT RULES

R U L E

VI PROMOTION OF WASTE TO ENERGY

- The SWM Rules, emphasize promotion of waste to energy plants. The rules mandate all industrial units using fuel and located within 100 km from a solid waste-based Refuse-Derived Fuel (RDF) plant to make arrangements within six months from the date of notification of these rules to replace at least 5 per cent of their fuel requirement by RDF so produced.
- Refuse-Derived Fuel (RDF) is a fuel (methane, methanol or ethanol) produced from various types of waste such as Municipal Solid Waste (MSW).
- As per the rules, the Ministry of New and Renewable Energy Sources should facilitate infrastructure creation for Waste to Energy plants and provide appropriate subsidy or incentives for such Waste to Energy plants.



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